

Practical Indoor Localization via Smartphone Sensor Data Fusion Techniques: A Performance Study

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Abstract—Accurate indoor localization constitutes a challenging yet fundamental research problem towards the large-scale deployment of next-generation mobile indoor location-based services. This paper addresses two key issues of indoor localization: (i) how to take benefit of the presence of inertial sensors, short-range and long-range radio interfaces on modern smartphones in order to achieve fine-grained localization and trajectory tracking, and -at the same time- (ii) how to perform it while limiting the impact on energy-constrained devices. To address the first issue, we propose a novel hybrid strategy which implements a dual-step fusion process, i.e. it merges the estimations produced by pattern matching algorithms applied to short-range and long-range wireless sources available on smartphones- and then it merges the estimations produced by Pedestrian Dead Reckoning (PDR) and Radio Fingerprinting (RF) techniques, in order to overcome the limitations of each approach. For the second issue, we describe the design and implementation of a novel client-server architecture, which offloads the computational expensive tasks to the infrastructure, while still guaranteeing acceptable localization lag. Finally, a modular, extensive evaluation is proposed on real-world scenarios, quantifying the impact of each sensor/source on the localization accuracy, and the gain induced by the dual-step fusion process over basic PDR localization techniques.

I. INTRODUCTION

According to a recent survey conducted by the Pew Research Center, accessing location-aware data and recommendations constitutes the most common usage pattern of smartphone by Americans, beside conventional services like voice calls and texting [1]. Such trend is not surprising if we consider the number and variety of Mobile Location-Based Services (MLBS) available on the market, for traditional (e.g. navigation, social-media platforms) and innovative application domains (e.g. crowdsensing) [2]. The current frontier of MLBS is represented by services for the indoor environments, including applications for residential and work locations, as well as novel context-aware systems enabled by the Internet of Things (IoT) [3]. Indoor MLBS can leverage on the hardware equipment of commercial off-the-shelf (COTS) smartphones, consisting of a wide range of embedded sensors, wireless radio interfaces and cameras. Although no reference solution exists on the market, the number of research studies and proof-of-concepts on Indoor Localization Systems (ILS) is significant, further demonstrating the relevance and timeliness of the topic. Excluding camera-based solutions, which are considered too costly and computationally expensive for most of general-purpose indoor applications, the existing ILS can be classified in three major categories: Pedestrian

Dead Reckoning (PDR) -based, Radio Fingerprinting (RF) -based, or Hybrid systems. PDR systems use inertial sensors (e.g. accelerometer/gyroscope) to estimate the step length and orientation, and from them the current location [4] [5] [6]. While these solutions do not require external infrastructures and hence can be implemented on smartphone as stand-alone software, they also experience run-to-run biases accumulating over time [7] [8]. Vice versa, RF-based solutions exploit the wireless signal diversity observed at different locations of the scenario; such diversity can be originated by the high density of Wi-Fi APs on urban environments or by the installation of ad-hoc, short-range transmitters, like BLE beacons [9]. Finally, hybrid localization systems combine the outputs of PDR and RF techniques while attempting to overcome the limitations of each: the two techniques can be executed concurrently -e.g. by setting the weights of a Particle Filters (PF) algorithm - or sequentially, e.g. the PDR location is adjusted through the RF estimation at reference scenario locations [10], [11] [12] [13], [14]. At the same time, we remark that most of the solutions proposed in the literature are not suitable for MLBS, since (i) they involve *computational-expensive* algorithms to be executed on energy-constrained devices and/or (ii) they do not take into account realistic *performance constraints* of COTS device, like the sensing delay, and/or (iii) they rely on complex *signal processing techniques* which might be not implementable through the APIs provided by the Mobile Operating System (MOS).

In this paper, we design a novel hybrid ILS for smartphones - called Wireless-LOCator (WI-LO)- that exploits multiple techniques, i.e. PDR, RF, Human Activity Recognition (HAR), and sources i.e. the Wi-Fi, LTE, BLE, the magnetometer (MAG) in order to achieve fine-grained indoor localization, while still providing a practical and energy-efficient solution for Android devices. The proposed study provides three main research contributions compared to the existing literature on ILS:

- 1) We describe a novel *fusion technique* for indoor localization, which combines the sensing data of inertial sensors (ACC/GYR), the magnetometer (MAG), the short-range radio interfaces (the Wi-Fi and the BLE) and long-range ones (e.g. the LTE). The fusion scheme is driven by the HAR module, and works in two steps: each time an idle activity has been detected, the selected radio interfaces (i.e. Wi-Fi, LTE, BLE, or MAG) are sampled, and a RF localization is performed. The output of the

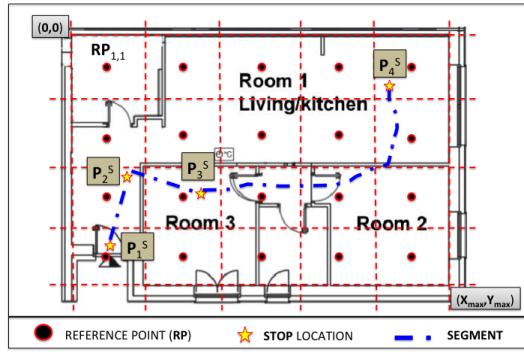


Fig. 1. Reference model of the indoor scenario.

RF scheme is then merged with the current output of the PDR, in order to mitigate the navigation errors caused by imprecise step detection.

- 2) We implement the scheme described above through a *client-server architecture*, which offloads computational extensive tasks (e.g. pattern matching schemes) on the infrastructure, while executing more lightweight tasks (e.g. PDR position update) on the smartphone. Moreover, differently from other studies, we propose a PDR/RF fusion technique that does not involve the utilization of statistical filters and iterative procedures, rather it uses proxies for the confidence of each localization technique.
- 3) We propose an *extensive, modular evaluation* of WI-LO, by investigating the contribution of each sensor on the overall localization accuracy, of the PDR and RF techniques taken in isolation, and of the hybrid PDR/RF system. The experimental results demonstrate that the proposed ILS is able to considerably improve the performance of PDR-only solutions under different environments and for different hardware devices.

In [16], we have presented a preliminary version of WI-LO based on the RF technique only. Here, we significantly extend the study, by adding the PDR scheme, the hybrid PDR/RF localization scheme, and a new performance evaluation.

The rest of the paper is structured as follows. The literature on ILS based on PDR and hybrid solutions is surveyed in Section II. The system model and proposed WI-LO architecture, components and algorithms, are discussed in Section III. Implementation and performance evaluation are presented in Section IV. Conclusions follow in Section V.

II. RELATED WORKS

We can classify the existing indoor localization techniques in two main categories, i.e.: (i) infrastructure-less ILS based on PDR techniques, and (ii) hybrid solutions, combining PDR and RF techniques. In the first case, the system is self-contained, however it must cope with the phone orientation and the carrying mode as well as with the imprecision of sensory data. About the first issue (i.e. device orientation), the authors of [4] propose a novel decomposition method,

aimed to identify the carrying pattern via Machine Learning (ML) classification algorithms. Regarding the navigation errors induced by run-to-run biases and imprecise step length detection, common solutions employ statistical filters (e.g. [5], [6]) or calibration techniques (e.g. [8]). More specifically, in [5] [6], methods for accelerometer sampling, step detection and turn action recognition are proposed and coupled with a Particle Filter (PF) based technique for map matching. A similar approach is investigated in [7], by employing both Kalman Filter (KF) and PF, respectively for floor and 2D position detection. An innovative calibration scheme is described in [8]: here, the indoor position -estimated via PDR- is adjusted by considering the relative distance from other mobile peers, whose arrival at specific scenario landmarks has been detected via audio ranging. Hybrid PDR/RF techniques can be further classified based on the fusion technique in use. The usage of PF is investigated in several recent studies, like [10], [11] and [12]: more specifically, the solution in [10] assigns a weight to each particle, which is a linear combination of the likelihood of five different inputs (including the PDR, the Wi-Fi and the distances from known reference points), while in [11] the weight is a function of the magnetometer likelihood. Map information are introduced in [12], so that particles violating physical constraints of the scenario (e.g. presence of walls) are filtered out. KF-based techniques are employed in [13] and [14], where the Wi-Fi RF is used to reduce the initial search space.

Differently from the studies described so far, the WI-LO system provides the following novel features: (i) it exploits all the available radio technologies of the smartphone for fine-grained indoor localization, by considering a confidence threshold for each technology; (ii) it relies on software libraries made available by the Android OS, hence it can be deployed on COTS smartphone devices without requiring modifications to the stack or violations of the MOS security mechanisms; (iii) it limits the energy overhead involved by the indoor localization and RF pattern matching process, thanks to the client-server architecture described in the Section below.

III. SYSTEM MODEL AND PROPOSED ARCHITECTURE

We consider a 2D indoor scenario like the one depicted in Figure 1. We assume a discrete scenario decomposition into a grid of square cells, with side equal to d meters: a Reference Point (RP) denotes the center of each cell. The RF scheme will return the RP corresponding to the estimated cell. Vice-versa, the PDR scheme performs continuous tracking of the user trajectory. This latter is composed by a set of segments: let $s_i = (P_i^s, P_i^e)$ be the i -th segment, and P_i^s, P_i^e respectively the starting point and end of the segment, with $P_i^e = P_{i+1}^s$. These points correspond to locations where the user stopped, and are exploited to enhance the PDR estimations, as better detailed in the following.

The WI-LO architecture is depicted in Figure 2. The client component is an Android app, which tracks the user trajectory via the PDR scheme detailed in Section III-A. Moreover, the app is in charge of sampling the values of the embedded

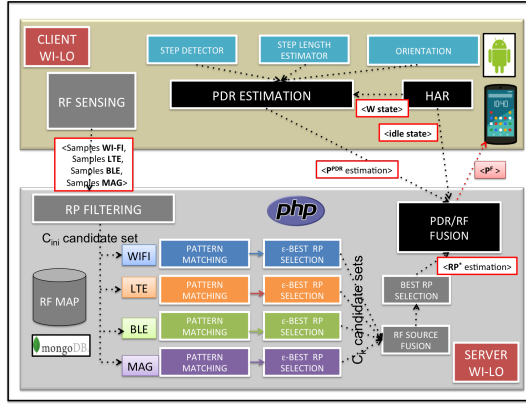


Fig. 2. The WI-LO client/server architecture.

accelerometer, and, based on them, to distinguish whether the user is currently in *idle* or in *walking* state. Leveraging on our HAR framework presented in [15], the mobile client can recognize complementary human activities, like events of stairs down-climbing or up-climbing; based on them, the current user floor can be detected with a reasonable accuracy [15]. However, we do not further elaborate on the issue on the current study, which is focused on 2D indoor localization. The proposed ILS works in three steps:

- 1) First, current position and HAR state are estimated via inertial sensor monitoring;
- 2) Second, whenever the *idle* state has been detected, discrete RF localization is performed. To this aim, the app gathers the RSS samples from the selected radio interfaces/sensors, i.e. the WiFi, LTE, BLE or MAG, and transmits them to the WI-LO server. Here, the Diversity-based fusion scheme described in Section III-C is executed.
- 3) Third, the final indoor location is computed by merging the RF and PDR estimations, according to the fusion rule described in Section III-D and implemented within the Fusion Module of Figure 2. Also, the starting point for the next segment is adjusted.

In the following, we detail each of the previous steps, which are evaluated separately in Section IV.

A. PDR-based Localization

The PDR scheme implements the navigation Equations below to update the user position at each detected step:

$$\begin{aligned} x(t+1)^i &= x(t)^i + D \cdot \sin(\theta) \\ y(t+1)^i &= y(t)^i + D \cdot \cos(\theta) \end{aligned} \quad (1)$$

Here, $x(t)^i, y(t)^i$ denotes the previous location within segment i , D is the step length while θ is the orientation angle. This latter is estimated via the digital compass (i.e. magnetometer and gyroscope); further details on the calibration phase are provided in Section IV. Equation 1 is implemented through the Step Detector (SD) and Step Length Estimator (SLE) modules of Figure 2. Regarding the step detection,

which is a well known issue of PDR-based ILS, the WI-LO app relies on the virtual step detector sensor available on recent Android OS versions. The SLE module provides two techniques of step length estimation. The first one, simply named Static Step Length (SSL), employs a well-know empirical method, i.e. it assumes a constant value for the step length SL , based on the user gender and height:

$$SL = \alpha \cdot h \quad (2)$$

where $\alpha=0.415$ for male users, $\alpha=0.413$ for female users. The second technique, named Dynamic Step Length (DSL) and described in [17], estimates the step length based on the minimum and maximum peaks registered by the accelerometer:

$$SL = k \cdot \sqrt[4]{A_{max} - A_{min}} \quad (3)$$

where k is a calibration factor. The accuracy of the SD module and the comparison between the SSL and the DSL techniques are investigated in Section IV.

B. Human Activity Recognition

The HAR Module is fully implemented on the mobile client, and is in charge of discriminating between the *idle* and *walking* state. Since the activities to be detected are well distinguishable, we do not employ learning-based techniques like most of HAR systems [15], rather we implement a simple yet effective threshold-based detector. Every T_a seconds, the values of the triaxial accelerometer, i.e. $a_x(t), a_y(t), a_z(t)$ are sampled through the Android OS API, and synthesized via the *magnitude* metric [15]:

$$m(t) = \sqrt[2]{a_x^2(t) + a_y^2(t) + a_z^2(t)} - g \quad (4)$$

where g is the acceleration gravity. Let Th_{idle} be a threshold on the magnitude value, set to 0.1 in our experiments and any case tunable through the WI-LO app. If $m(t) > Th_{idle}$, then the current state is set as *walking*, and the current PDR estimation is returned to the user. Otherwise, the current state is set to *idle*, and the RF localization and then the PDR/RF fusion algorithm are executed.

C. RF-based Localization

We assume that the RF training phase has already been performed, by following the procedure detailed in [16]. Each time the WI-LO client is in *idle* mode, it senses the environment for T_s seconds, by using any combination of the Wi-Fi, LTE, BLE and MAG sources. The source selection can be configured by users through the WI-LO GUI of the mobile app. For each detected Wi-Fi AP and BLE beacon, the following record is extracted:

$$R_j^{off} = \langle BSSID_j^{off}, RSS_j^{off}, \mu_j^{off}, \sigma_j^{off} \rangle \quad (5)$$

where $BSSID_j^{off}$ is the MAC address of AP j , RSS_j^{off} is the set of RSS samples, μ_j^{off} and σ_j^{off} are respectively the mean and variance RSS value computed over RSS_j^{off} . For the LTE and MAG measurements, a single record is built without the BSSID field. The gathered records are then transferred

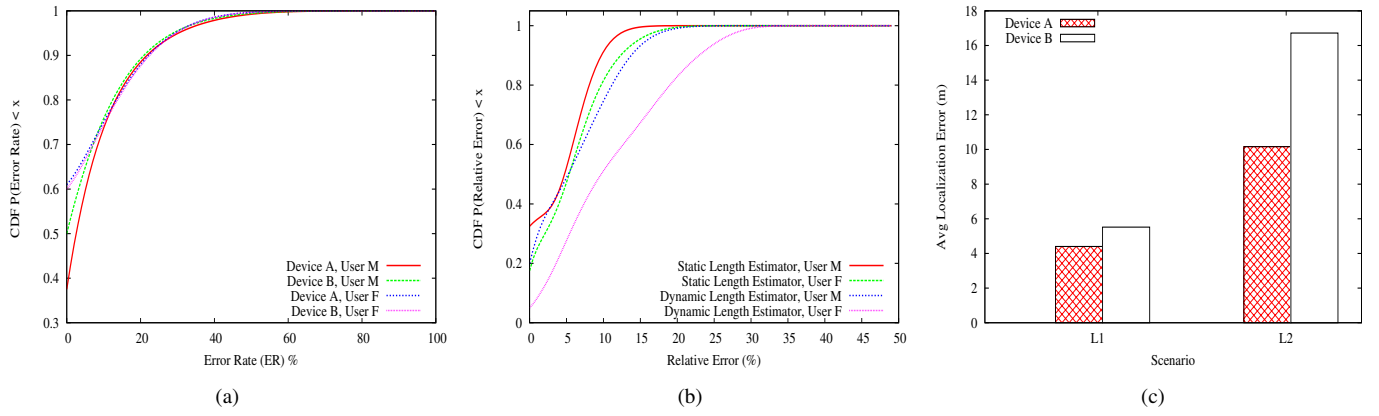


Fig. 3. The CDFs for the Error Rate (Step Detection analysis) and the Relative Distance Error (Step Length Estimator analysis) are depicted in Figures 3(a) and 3(b), respectively. The average localization error for the PDR-only technique is depicted in Figure 3(c).

to the WI-LO server, where the estimated RP (denoted as RP^*) is computed and passed as input to the PDR/RF Fusion module. The RF localization process follows three main steps:

- 1) *Filtering*. We restrict the set of RP candidates to those which are within a range of R_{thr} meters from the PDR location estimation. Let C_{ini} be such set. Moreover, let I_{on} denote the set of selected sources.
- 2) *Matching*. On each source, we apply a Pattern Matching (PM) algorithm among the ones implemented in WI-LO, and including statistical techniques (e.g. the Gaussian Likelihood) as well as ML techniques. A detailed description of the PM schemes available in WI-LO can be found in [16]; we just remark here that, differently from other frameworks (e.g. [10] [12]), we do not merge samples of different sources at this stage. For each source k , a set of candidate locations $C_k \subset C_{ini}$ is identified. Moreover, a score value $S_k(i) \in [0 : 1]$ is computed for each $RP_i \in C_k$.
- 3) *RF Fusion*. Finally, we assign a Fusion Score to each RP $i \in C_{ini}$, as follows:

$$FS(i) = \sum_{k=1}^{|I_{on}|} U_k(i) \cdot \omega_k(i) \quad (6)$$

where $U_k(i)$ is the identity function returning 1 if RP $i \in C_k$, 0 otherwise, and $0 \leq \omega_k(i) \leq 1$ is a weight for source k . The RP j associated to the maximum value of FS , i.e. $j = \text{argmax} FS(j)$ represents the output of the RF localization scheme.

In [16], we proposed and compared different algorithms for the computation of the source weights $\omega_k(*)$. Here, we mainly focus on the Diversity-based fusion scheme that aims to identify location- and source-dependent signal anomalies, which can work as unique fingerprints of each specific RP. To this purpose, we compute a Diversity Metric value (DM) between each couple of RPs of the scenario and for each source. The $DM_k(i, j)$ value between RP i and j on source

k is defined as follows:

$$DM_k(i, j) = \frac{|S_k(i) - S_k(j)|}{\max(S_k(i), S_k(j))} \quad (7)$$

where $S_k(i)$ is the score function applied by the selected PM algorithm on source k and for RP i . The DM metric: (i) is agnostic of the specific PM algorithm in use, (ii) takes values in range $[0, 1]$. We then define the Diversity Metric of RP i on source k , i.e. $DM_k(i)$, as the average $DM(i, j)$ considering all the other RPs j of the scenario, i.e.:

$$DM_k(i) = \frac{1}{N_{rp}} \sum_{j=1, i \neq j}^{N_{rp}} DM_k(i, j) \quad (8)$$

where N_{rp} is the number of RPs available in the scenario. A high value of $DM_k(i)$ indicates that source k is able to recognize RP i with high precision. Then, we compare the DM values of different sources, and we assign the $\omega_k(i)$ values proportionally to $DM_k(i)$, i.e.:

$$\omega_k(i) = \frac{DM_k(i)}{\sum_{l=1}^{|I_{on}|} DM_l(i)} \quad (9)$$

The $\omega_k(i)$ values are pre-computed at the end of the training phase, and stored within the database on the WI-LO server.

D. PDR/RF Fusion rule

Let P_i^s be the starting point of the current segment, i.e. the latest position where a PDR calibration/fusion has been performed. Let $P^{PDR} = \langle x^{PDR}, y^{PDR} \rangle$ be the current estimation computed by the PDR technique, and $RP^* = \langle x^{RP^*}, y^{RP^*} \rangle$ the estimation computed by the RF technique. The Fusion Module of WI-LO computes the final location x^F, y^F , which is then returned to the client and displayed on the mobile app. To this purpose, the fusion rule computes a weighted average of the PDR/RF estimations, i.e.:

$$\begin{aligned} x^F &= x^{PDR} \cdot (1 - \beta) + \beta \cdot x^{RP^*} \\ y^F &= y^{PDR} \cdot (1 - \beta) + \beta \cdot y^{RP^*} \end{aligned} \quad (10)$$

The β factor is formulated as follows:

$$\beta = (1 - \frac{D(RP^*, P_i^s)}{D(RP^*, P_i^s) + D(P^{PDR}, P_i^s)}) \cdot \max_k DM_k(RP^*) \quad (11)$$

Here, $D(\cdot, \cdot)$ denotes the Euclidean distance between 2D points. Hence, $D(RP^*, P_i^s)$ is the distance between the current RP and the previous user location, while $D(P^{PDR}, P_i^s)$ is the walked distance since the last data fusion. In Equation 11, we give more credit to the PDR estimation, i.e. the β factor is high, when: (i) the current spatial granularity of the RF technique is higher than the PDR granularity, e.g. a long walk has been completed without stops and (ii) the RF localization has high confidence regarding the current RP, since its fingerprint is significantly different from other RPs.

IV. IMPLEMENTATION AND PERFORMANCE EVALUATION

In this Section, we show a subset of the experimental performance results of the proposed ILS. We implemented the client WI-LO on Android OS (version ≥ 5.0). The mobile app can be used in three modes, i.e. *mapping*, *training* and *testing*. In *mapping* mode, the user can define the characteristics of a multi-floor building object, including names and locations of the Reference Points (RPs). In *training* mode, the user populates the RF maps of the environment, simply moving at each RP, and selecting the combination of sources to use for sensing purposes (Wi-Fi, BLE, LTE, MAG). The building data are then transferred to the WI-LO backend; in this way, multiple users can share the effort of mapping a new environment, or can self-position in a new building, leveraging on the training data produced by other users. Finally, in *testing* mode, the user selects a building/floor (previously mapped and trained) and the proper localization technique, choosing among: PDR only, RF only or Hybrid. A preference panel allows the user selecting the RF sources and algorithms, including pattern matching and fusion rules [16]. The WI-LO back-end has been implemented in PHP and Mongo-DB, and communicates with the client through HTTP REST.

Testing methodology. A well-known issue of smartphone-based ILS is the remarkable impact of different hardware devices on the localization performance. To this purpose, we used two different devices in the tests: a LG Nexus 5 (simply called as Device A in the following) and a Oneplus X (Device B). Similarly, we aggregated the results based on the user gender (M=male, F=female). We repeated the tests at two different indoor locations within the UNIBO-DISI department: (i) the basement floor (simply called as L1 in the following) and (ii) the ground floor (L2 in the following). We propose a modular evaluation of the proposed ILS. First, in Section IV-A, we evaluate the performance of the PDR-only localization system, further characterizing the impact of the SD and SLE components described in Section III-A. In Section IV-B, we evaluate the discrete localization at RPs via the RF source fusion scheme described in Section III-C. Finally, in Section IV-C we discuss the performance gain produced by the Hybrid PDR+RF scheme of Section III-D.

A. PDR-only localization

First, we evaluate the accuracy of the Step Detector module of the WI-LO client. On the x-axis of Figure 3(a), we show the Error Rate (i.e. the rate of step mis-detection), while the y-axis is the Cumulative Distribution Function (CDF) for the same index. We plot different lines according to users (M/F) and devices (A/B) involved in the tests. No appreciable difference can be observed when varying the device type; moreover, we can notice that the ER is zero (i.e. perfect step detection) in 60% of the cases for F users, and is lower than 10% in 75% of the cases for both M/F users. Clearly, higher detection accuracy can be achieved by implementing fine-grained signal processing techniques on the raw accelerometer values. However, the current solution constitutes a reasonable trade-off between performance accuracy and implementation complexity. Figure 3(b) evaluates the performance of the second component of Equation 1, i.e. the Step Length Estimator, when using the SL or the DL methods described in Section III-A. We considered a straight path of 10 steps, and we measured the difference between the real and estimated distance. We show the Relative Error (%) on the x-axis, and the CDF of the same index on the y-axis. We can notice that the SL method is more accurate than the DL, for both users; for this reason, we adopted the SL method in the following. Finally, we evaluated the PDR with all the components (SD, SLE and orientation) enabled, when tracking non-straight pedestrian trajectories on scenarios L1 and L2. Figure 3(c) shows the average localization error for the two devices. We found that: (i) the localization error is lower for Device B, but in both cases is higher than 4 meters; (ii) the trajectory is considerably imprecise for the L2 scenario, mainly due to the magnetic disturbance caused by the reinforced concrete walls.

B. RF-only localization

We consider the environment L1, characterized by an high density of Wi-Fi APs (> 15 on each RP). On Figure 4(a) we show the RF localization accuracy, i.e. the ability to detect the reference cell/RP on which the user is located, for $d=2m$, and two configurations of the sensing time, i.e. $T_s=2sec$ and $T_s=5sec$, while varying the RF source in use. In all cases, we used the Nearest Neighbour (NN) pattern matching since this algorithm has been shown to achieve the highest accuracy among the ones available in WI-LO [16]. The Majority scheme is considered as the baseline source fusion algorithm: it executes the NN algorithm on each data source k , and returns the candidate RP^* occurring more in the candidate set $C = \bigcup_k C_k$. We can notice that: (i) Wi-Fi is the most accurate source, thanks to the high density of Wi-Fi APs in L1; (ii) the LTE and MAG sources do not guarantee -on average- high accuracy, hence applying a simple Majority fusion rule does not provide any benefit compared to the Wi-Fi only scheme; (iii) however, the combined utilization of LTE and MAG can help reducing the localization errors at some specific RPs, where the LTE and MAG signals present significant diversities: this explains the additional gain provided by the Diversity scheme over the Wi-Fi only technique. Reducing

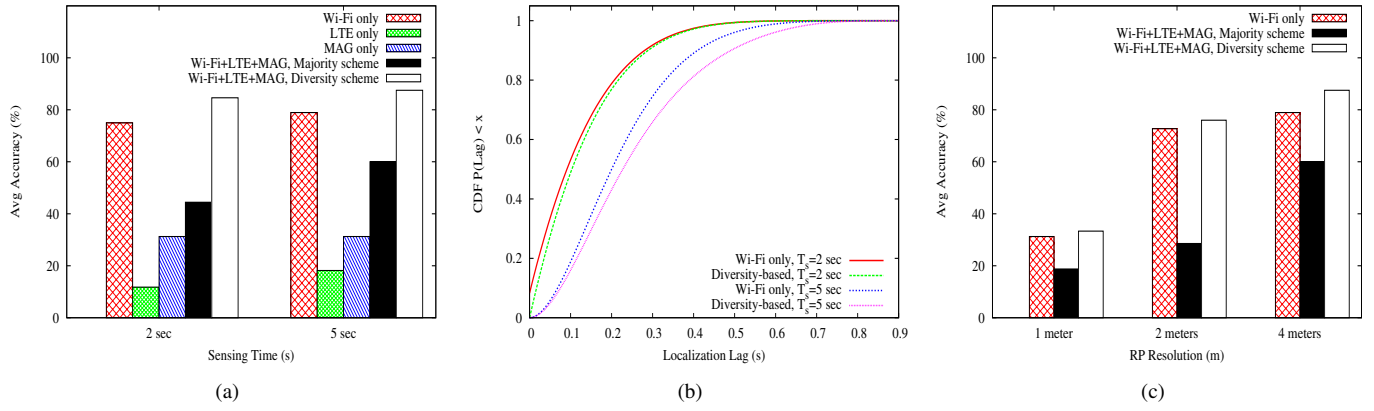


Fig. 4. The localization accuracy of the RF-only technique for different values of the sensing time (T_s) and of the RP resolution (d) are shown in Figures 4(a) and 4(c), respectively. The CDF of the localization lag is reported in Figure 4(b).

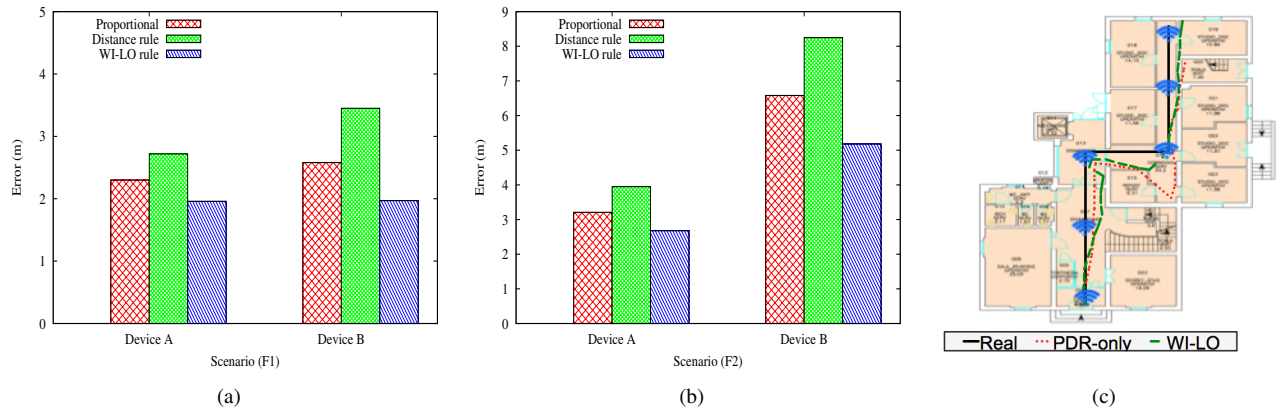


Fig. 5. The localization error for the L1 and L2 scenarios is depicted in Figure 5(a) and 5(b), respectively. An example of trajectory computed by the PDR-only and WI-LO fusion rule for the L1 scenario is shown in Figure 5(c).

the sensing time, the performance of all schemes become slightly worse; however, the accuracy of the Diversity-based scheme is still higher than 85%. Figure 4(b) provides further insight on this issue (i.e. the stop duration), showing CDF of the localization lag, described as the overhead induced by the localization task. Such lag includes the computation time at the WI-LO backend and the communication time between client/server, and must be summed to the sensing time T_s previously mentioned. The source fusion schemes (i.e. Majority and Diversity-based) require additional computation because the pattern matching algorithm is executed on each source; this translates into higher lag values compared to the Wi-Fi only solution. However, in any case, the lag of the Diversity-based scheme does not exceed the 0.7 seconds in the worst cases (0.3 sec is the mean value), and is always lower than the sensing time T_s . Finally, in Figure 4(b) we investigate the spatial resolution of RF-based techniques, for three different configurations of d . When decreasing d , the training time increases significantly, and the pattern matching algorithms are asked to discriminate between closer RPs with similar fingerprints. We can see that, for $d = 1$, the accuracy is lower than 40%, and the Diversity and Wi-Fi only schemes

perform similarly, since the utilization of LTE and MAG sources does not provide any benefit. The accuracy of all the schemes increases when $d \geq 2$, which constitutes the minimal spatial resolution of the RF technique in the L1 scenario.

C. Hybrid localization

In this Section, we evaluate the performance of the full WI-LO system implementing hybrid PDR/RF localization. For the RF, we consider the Diversity scheme with $T_s=5\text{sec}$ since the previous results demonstrated that this is the most accurate RF technique on the testing scenarios. In these experiments, we assume that the user will stop at the RPs marked with the Wi-Fi icon on Figures 5(c) and 6, and adjust the starting point of the next segment by merging the outputs of PDR and RF localization techniques. We compare three fusion rules, i.e.:

- *Proportional*. This method simply averages the estimations produced by PDR and RF, by setting $\beta=0.5$.
- *Distance rule*. This method [18] computes a weighted average of the estimations, setting β as follows:

$$\beta = \frac{\frac{1}{\text{dist}_{\text{wifi}}}}{\frac{1}{\text{dist}_{\text{wifi}}} + \frac{1}{\text{dist}_{\text{pdr}}}} \quad (12)$$

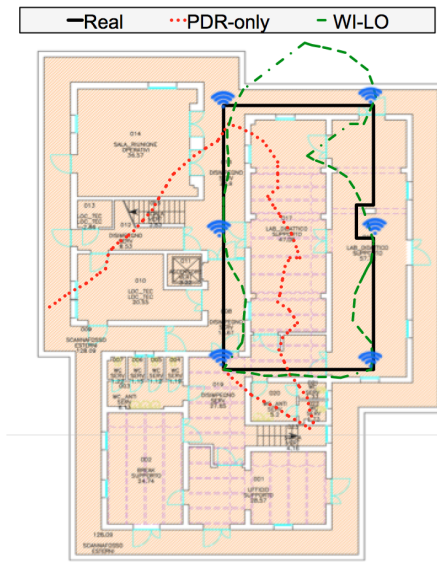


Fig. 6. An example of trajectory computed by the PDR-only and WI-LO fusion rule for the L2 scenario is shown in Figure 6.

where $dist_{wifl}$ and $dist_{pdr}$ are the distance between from the previous and current location computed through RF and PDR techniques respectively.

- **Wi-LO rule.** This method implements the fusion rule presented in Section III-D.

Figure 5(a) depicts the average localization error on the L1 scenario, for the algorithms defined above and the two devices. We can notice that: (i) all three Hybrid schemes achieve significant improvement compared to the PDR-only solution; (ii) the WI-LO fusion rule provides the best performance, by reducing the localization error of more than 25% compared to the Proportional scheme, and of more than 50% compared to the Distance-based; (iii) the same trends can be observed on devices A and B. In Figure 5(b), we repeated the same test but for the L2 scenario. Here, the PDR-only localization was highly inaccurate due to the magnetic disturbance on the underground floor (see Figure 3(c)). Although worse than L1, the performance of hybrid PDR/RF schemes are in any case considerably better than the PDR-only scheme. Moreover, the WI-LO rule again minimizes the navigation errors for both devices. Finally, in Figures 5(c) and 6, we plot some trajectories derived from the localization tests, respectively in the L1 and L2 scenario. The real trajectory is denoted by the black line; the other two lines represent the trajectories computed by the PDR-only and by the Hybrid ILS.

V. CONCLUSIONS AND FUTURE WORKS

In this paper, we have presented WI-LO, a novel framework supporting the deployment of next-generation location based services for indoor environments. The WI-LO platform provides fine-grained indoor localization of smartphone devices by employing a great variety of sensors available on modern smartphones, including embedded sensors (ACC, GYR,

MAG) and radio communication interfaces (Wi-Fi, LTE, BLE), and multiple sensor data analytics techniques, including Human Activity Recognition (HAR), Radio Fingerprinting (RF) and Pedestrian Dead Recognition (PDR). Future works include -on the framework implementation side- the possibility for users to import 2D planimetry on multiple formats, and to share the localization data with other apps. On the algorithmic side, we plan to investigate techniques to reduce the overhead of the training phase, i.e. populating the RF maps while users will move within the scenario via trajectory matching and crowdsensing techniques. On the evaluation side, we plan to test our framework on multi-floor building scenarios.

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